[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #4

ATK^2 :-

Quadratic Equation

$\boxed{D=2}$ '='

$(x^1-2)^2 = 1$ Quad. ??

$\Rightarrow (x)^2 + (2)^2 - 2(x)(2) = 1$

$\Rightarrow \underbrace{x^2} + \underbrace{4} - \underbrace{4x} = 1$

$\Rightarrow x^2 - 4x + 4 - 1 = 0$

$\Rightarrow \boxed{x^2 - 4x + 3 = 0}$

D = 2 Quad ✓✓

$\boxed{ax^2 + bx + c = 0}$

solve ← x की value

① ~~Middle term splitting~~ ✓

उलझे sawaal

② ~~Discriminant Method.~~ ✓

#K³B

Solve ⇒ $1x^2 + 1x - 2 = 0$ ← 3 terms

$\underbrace{x^2 + 2x} - \underbrace{1x - 2} = 0$ ← 4 terms

$x(x+2) - 1(x+2) = 0$ ← common में लिया

⋮

अगर directly 4 terms

ऐसे ⟶ common लेलो

#LP: Solve the following quadratic equations by factorization method:

$\Rightarrow 4x^2 - 2(a^2 + b^2)x + a^2b^2 = 0$ [CBSE 2004]

$\Rightarrow \underbrace{4x^2 - 2a^2x}\ \underbrace{-2b^2x + a^2b^2} = 0 \leftarrow$ 4 terms

common लेलो

$\Rightarrow 2x(2x - a^2) - b^2(2x - a^2) = 0$

$\Rightarrow (2x - a^2)[2x - b^2] = 0$

$2x - a^2 = 0$	$2x - b^2 = 0$
$2x = a^2$	$2x = b^2$
$x = a^2/2$ ✓	$x = b^2/2$ ✓

$$\#LP:\ \underbrace{a^2b^2x^2 + b^2x}-\underbrace{a^2x-1}=0 \longleftarrow \text{4 terms}$$

common
लेलो

$$b^2x\left(a^2x+1\right)-1\left(a^2x+1\right)=0$$

$$\left(a^2x+1\right)\left[b^2x-1\right]=0$$

$$a^2x+1=0 \qquad\qquad b^2x-1=0$$

$$x=-\frac{1}{a^2}\ \checkmark \qquad\qquad x=\frac{1}{b^2}\ \checkmark$$

#LP: $\dfrac{1}{a+b+x} = \dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{x}, (a+b \neq 0)$

$$\Rightarrow \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\text{LCM} \Rightarrow \frac{1(x) - 1(a+b+x)}{(a+b+x)(x)} = \frac{1(b)+1(a)}{ab}$$

$$\Rightarrow \frac{\cancel{x} - a - b - \cancel{x}}{(ax+bx+x^2)} = \frac{a+b}{ab}$$

$$\Rightarrow \frac{-a-b}{(ax+bx+x^2)} = \frac{a+b}{ab}$$

$$\Rightarrow \frac{-\cancel{(a+b)}^{1}}{(ax+bx+x^2)} = \frac{\cancel{a+b}^{1}}{ab}$$

$$\Rightarrow -ab = ax+bx+x^2$$

$$\Rightarrow \boxed{x^2 + ax + bx + ab = 0}$$

Common लेलो

$$\Rightarrow x(x+a) + b(x+a) = 0$$

$$\Rightarrow (x+a)[x+b] = 0$$

$x+a=0$ | $x+b=0$

$x=-a$ ✓ | $x=-b$ ✓

#Discriminant Method: / श्री धराचार्य Method

$$ax^2+bx+c=0$$

$(b^2-4ac) =$ Discriminant (D)

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-b \pm \sqrt{D}}{2a}$$

D

$$x = \frac{-b+\sqrt{b^2-4ac}}{2a} \qquad x = \frac{-b-\sqrt{b^2-4ac}}{2a}$$

#LP: Write the discriminant of the following quadratic equations and hence find the roots:

(i) $2x^2 - 5x + 3 = 0$

$ax^2 + bx + c = 0$

$D = b^2 - 4ac \Rightarrow (-5)^2 - 4(2)(3)$

$\Rightarrow 25 - 24$

$\boxed{D = 1}$

Roots

$x = \frac{-b \pm \sqrt{D}}{2a}$

$= \frac{-(-5) \pm \sqrt{1}}{2(2)}$

$\Rightarrow \frac{5 \pm 1}{4}$

$\frac{5+1}{4}, \frac{5-1}{4}$

$\Rightarrow \frac{\cancel{6}^{3}}{\cancel{4}_{2}}, \frac{\cancel{4}^{1}}{\cancel{4}}$

$\Rightarrow \frac{3}{2}, 1$ x

(ii) $x^2 + 2x + 4 = 0$

$D = b^2 - 4ac \Rightarrow (2)^2 - 4(1)(4)$

$\Rightarrow 4 - 16$

$\boxed{D \Rightarrow -12}$ ← -ve

No real roots

Roots

$x = \frac{-b \pm \sqrt{D}}{2a}$

$= \frac{-(2) \pm \sqrt{-12}}{2(1)}$

Root के अंदर -ve

Impossible!!

NATURE OF ROOTS

$\boxed{D} \rightarrow$ Powerful cheez !!

अलग-अलग

$D = b^2 - 4ac$

- $D > 0 \leftarrow$ Real and Distinct Roots
- $D = 0 \leftarrow$ Real and Equal roots
- $D < 0 \leftarrow$ Unreal roots

#LP: Determine the nature of the roots of the following quadratic equations: D

(i) $2x^2 - 6x + 3 = 0$

(ii) $(3/5)x^2 - (2/3) + 1 = 0$

(iii) $3x^2 - 4\sqrt{3}x + 4 = 0$ HW

(iv) $3x^2 - 2\sqrt{6}x + 2 = 0$

(i) $D = (-6)^2 - 4(2)(3)$

$= 36 - 24$

$D = 12$

$D > 0 \leftarrow$ real & distinct roots

(iv) $D = (-2\sqrt{6})^2 - 4(3)(2)$

$= (-2)^2(\sqrt{6})^2 - 24$

$\Rightarrow 4(6) - 24$

$D = 24 - 24$

$\boxed{D = 0}$

real & equal roots

#LP:If the roots of quadratic equation $\boxed{4x^2 - 5x + k = 0}$ are real and equal, then the value of k is (CBSE 2024)

(a) $\frac{5}{4}$

(b) $\frac{25}{16}$ 49%

(c) $-\frac{5}{4}$

(d) $-\frac{25}{16}$

$$D = 0$$

$$b^2 - 4ac = 0$$

$$\Rightarrow (-5)^2 - 4(4)(k) = 0$$

$$\Rightarrow 25 - 16k = 0$$

$$\Rightarrow 25 = 16k$$

$$\Rightarrow \boxed{k = \frac{25}{16}}$$

#LP:If the roots of equation $\boxed{ax^2 + bx + c = 0}$, $a \neq 0$ are real and equal, then which of the following relations is true? (CBSE 2024)

(a) $a = b^2/c$

(b) $b^2 = ac$

(c) $ac = b^2/4$ 57%

(d) $c = b^2/c$

$D = 0$

$b^2 - 4ac = 0$

$b^2 = 4ac$

$$\boxed{\frac{b^2}{4} = ac}$$

1(P I) → complete

1 P(II) 5, 6, 7, 8

THANK YOU COODIESS !!